MINI PROJECT REPORT
MCA 3RD SEMESTER (2025 BATCH)

PAWCONNECT
An AI-Powered Centralized Pet Adoption Platform

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

SUBMITTED BY:

1. ADEL ABDULLA (Roll No: 2), Batch B
2. ANIRUDH SUGATHAN (Roll No: 18), Batch B

PROJECT GUIDE:
Miss. SHERNA MOHAN
School of Computer Applications

1. INTRODUCTION

PawConnect is a next-generation web application designed to fix the communication and management issues currently plaguing pet adoption. In today's cities, the process is often chaotic and scattered and lacks the trust needed for safe adoptions. PawConnect creates a centralized, secure hub for pet owners, shelters, and potential adopters. Using modern technology like React/Flutter for the frontend and a solid Spring Boot backend, the platform provides a

streamlined experience. This document details the functional needs, system boundaries, and design choices that will guide the development of the PawConnect project.

2. PROBLEM STATEMENT

Right now, pet adoption happens across a messy mix of animal shelters, rescue groups, social media pages, classified sites, and various standalone apps. This current landscape is disjointed and lacks transparency. Adopters struggle to find healthy, suitable pets in their area, while shelters and owners find it hard to connect with serious, reliable adopters.

Because there is no central hub, users have to jump between different networks, rely on insecure communication like personal calls or emails, and verify health and authenticity on their own. This lack of standards leads to fake listings, duplicate posts, and poor communication, which unfortunately causes animals to stay in shelters longer and contributes to pet abandonment. There is an urgent need for a trusted, standardized, and modern tech solution that can manage the entire adoption process from start to finish.

3. RELEVANCE OF THE PROJECT

PawConnect is an all-in-one web application meant to simplify and automate the entire adoption cycle. The system includes user profile management, verified listing workflows, AI-powered features, and real-time transaction tools. Key technical features include JWT-based security for different roles, organized listing tools, interactive interfaces, and strong admin controls.

To improve matching, the system uses intelligent recommendation engines based on user preferences, automated computer vision to identify breeds from photos, and ML-based rules to detect fake listings. It also includes secure, built-in instant messaging so users can avoid using insecure personal contact channels.

Out of Scope: The project does not include third-party payment processing (adoptions remain non-commercial), physical animal transport or logistics, integration with state-run veterinary databases, or migration of historical data from existing shelter systems.

4. OBJECTIVES

The main goal of PawConnect is to build a secure, intelligent digital marketplace that increases successful adoptions and supports animal welfare. Our specific goals are:

- **Centralize Adoption:** Create one unified platform for owners, shelters, and adopters to stop the current fragmentation.
- **Build Trust:** Prevent fraud through automated AI checks and central administrative vetting.

- **Streamline Communication:** Include an instant messaging system with chat history to speed up the adoption process.
- **Data-Driven Matching:** Use user profile tracking and preferences to recommend the most compatible pets to adopters.

5. SYSTEM ANALYSIS

Introduction

System analysis involves a detailed look at how the current adoption process works, identifying its flaws, and explaining how the proposed PawConnect application fixes them.

6. EXISTING SYSTEM

Currently, the ecosystem is fragmented. Shelters often rely on paper records or local spreadsheets. Adoption listings are usually just random posts on platforms like Facebook, Instagram, or broad classified sites (e.g., OLX, Craigslist). Adopters are left to search through this mess, manually tracking updates, verifying information, and communicating through unmonitored personal channels.

7. LIMITATIONS OF EXISTING SYSTEM

Current methods have several major operational weaknesses:

1. **Extreme Fragmentation:** There is no central database, forcing users to repeatedly check dozens of unrelated websites and groups.
2. **No Vetting:** Anyone can post a listing, creating a dangerous space for scams, stolen pets, and financial extortion.
3. **Inaccurate Data:** Medical histories, vaccination records, and ages are often missing, incomplete, or completely unverified.
4. **Poor Communication:** Relying on generic emails or personal phone calls causes delays, often leading to missed opportunities.
5. **No Smart Matching:** Platforms only provide static, chronologically sorted lists, making adopters sift through thousands of irrelevant records.

8. PROPOSED SYSTEM

PawConnect is a cloud-ready web application designed to bring all adoption stakeholders together. It uses a single-user architecture with admin overrides, allowing verified users to easily act as both adopters and listers—similar to modern peer-to-peer marketplaces.

The system uses a Spring Boot service grid connected to a MySQL database, paired with a modern, responsive frontend dashboard. PawConnect differentiates itself by embedding AI directly into the workflow: computer vision automatically identifies breeds from images, a classification layer flags potential fraud, and an intelligent engine personalizes dashboards based on user criteria.

9. ADVANTAGES OF PROPOSED SYSTEM

PawConnect offers clear improvements over the legacy model:

- **Automated Trust:** Double-layer protection using AI fraud detection and final manual admin review ensures listings are legitimate.
- **Smart Breed Identification:** Image processing algorithms automatically classify breeds upon upload, reducing manual entry errors.
- **Secure Interaction:** Integrated real-time messaging keeps all communication safe and organized.
- **Transparency:** Structured tracking for vaccinations, vet checks, and behavioral history ensures complete data.
- **Unified Account:** Users can adopt or list animals from a single profile without switching accounts.

10. FEASIBILITY STUDY

We conducted a feasibility study across technical, operational, and economic criteria to ensure this project is realistic for the MCA mini-project framework.

10.1 Technical Feasibility

We assessed the hardware, software, and development tools required for the project. The application uses the **React** framework for the frontend, **Node.js with Express.js** for the backend, and **MongoDB** as the database. These technologies are open-source, widely adopted, well-documented, and supported by a large developer community, making them reliable for building a scalable web application. AI-based features, such as pet breed recognition and personalized pet recommendations, will be developed using lightweight **Python-based machine learning services** that can integrate seamlessly with the Node.js backend through APIs. This technology stack is suitable for the project's scope and can be implemented effectively within the available time and resources.

10.2 Operational Feasibility

We evaluated how the system fits into daily routines. PawConnect uses a dashboard style familiar to users of sites like OLX, making it intuitive to navigate. Administrators have a clear control panel to monitor reports and verify listings. No complex training is needed, which ensures a smooth rollout.

10.3 Economic Feasibility

We analyzed the costs and benefits of developing PawConnect. The project is built entirely using **open-source technologies**, including **React**, **Node.js**, **Express.js**, **MongoDB**, and **Python** for AI-based features. Since these technologies are freely available, there are no software licensing costs. Development is carried out using existing hardware and freely available development tools, which minimizes overall expenses. The system's benefits, such as simplifying the pet adoption process, reducing manual effort, improving user experience, and assisting shelters through AI-powered features, outweigh the development costs, making the project economically feasible.

11. SOFTWARE ENGINEERING PARADIGM APPLIED

We are following an Agile framework, specifically a mix of Scrum and Iterative Prototyping. We chose this over a rigid Waterfall model because the AI features (like breed recognition) require ongoing testing and adjustment.

Development is split into three sprints:

1. **Sprint 1:** Data schemas, user authentication, and basic CRUD features for pet listings.
2. **Sprint 2:** Adoption workflow engine, messaging system, and admin interface.
3. **Sprint 3:** AI model integration, search optimization, and final integration testing. This approach keeps core functions stable while allowing complex modules to evolve based on feedback.

12. SYSTEM ENVIRONMENT

Introduction

This section defines the hardware and software required to build, host, and run the PawConnect application.

12.1 Software Requirements Specification

• Client OS: Modern browsers (Chrome 100+, Firefox, Safari)

• Server OS: Linux (Ubuntu 20.04+) or Windows 10/11

• Database: MySQL Community Server 8.0+

• Backend: Java JDK 17, Spring Boot 3.x with Spring Security

• Frontend: React JS or Flutter Web SDK

12.2 Hardware Requirements Specification

• Workstation: Intel i5 / AMD Ryzen 5 (Minimum 4 Cores)

• Memory: Minimum 8 GB RAM (16 GB suggested for local services)

• Storage: Minimum 20 GB free space

• Deployment: Shared cloud instance (e.g., AWS EC2 t3.medium or equivalent)

13. USER CHARACTERISTICS

The user base is split into two primary groups:

1. General User (Adopters, Owners, Shelters): People looking to adopt, give up a pet for adoption, or manage shelter listings. They need basic web literacy. The app provides a unified dashboard for browsing, listing, requesting, and chatting without requiring technical expertise.

2. Platform Administrator (Admin): Operators responsible for platform security and trust. They monitor dashboards, review AI fraud alerts, verify credentials, and handle reports. They require a stronger technical background and training in platform policies.

14. FUNCTIONAL REQUIREMENTS

This section outlines the key functional modules that serve as the foundation for the PawConnect platform.

14.1 Manage user Authentication Introduction:

- Manages user registration and login, verifies user identity, and provides secure access to the application based on user roles.

Inputs:

- User registration details (Name, Email, Password) and login credentials (Email and Password).

Processing:

- Validates user information, authenticates login credentials, identifies the user role (User or Admin), and grants appropriate access to the system.

Outputs:

- Successful login, secure access to the appropriate dashboard, or an error message if the credentials are invalid.

14.2 Manage user profile

• Introduction: Manages user profiles, account settings, pet listings, and adoption records.

• Inputs: User profile details, account updates, pet listing information, and adoption requests.

• Processing: Updates user information, manages personal pet listings, and maintains adoption history.

• Outputs: Updated user profiles, account information, pet listing records, and adoption history.

14.3 Manage pet listing

• Introduction: Manages pet listings by allowing users to add, update, view, and remove pets available for adoption.

• Inputs: Pet details, images, and listing updates.

• Processing: Stores, updates, and manages pet information while maintaining the status of each listing.

• Outputs: Published pet listings, updated pet information, and current adoption status.

Search and Discover pets

• Introduction: Enables users to search and browse pets available for adoption using search and filtering options.

• Inputs: Search keywords and filter criteria such as pet type, breed, location, age, and gender.

• Processing: Searches available pet listings and filters results based on the selected criteria..

• Outputs: A list of matching pet listings with relevant details.

Manage pet adoption

• Introduction: Manages the adoption process by allowing users to submit, review, and track pet adoption requests.

• Inputs: Adoption requests and owner decisions (accept or reject).

• Processing: Records adoption requests, updates their status, and manages the adoption process.

• Outputs: Updated adoption request status and completed adoption records.

Manage user communication

• Introduction : communication between pet owners and adopters during the adoption process.

• Inputs: Messages sent by users.

• Processing: Stores, delivers, and maintains conversations between users.

• Outputs: Chat conversations, message notifications, and message history.

Manage Platform Administration

• Introduction: Manages the overall platform by monitoring users, pet listings, and reported activities.

• Inputs: User information, pet listings, reports, and administrative actions.

• Processing: Reviews user accounts and pet listings, manages reported content, and oversees platform activities..

• Outputs: Updated user accounts, approved or removed pet listings, resolved reports, and platform summary information.